

HR Employee Attrition Analytics Project

A Complete HR Analytics Study Using Python & Power BI

Python

BI

Statistics

DAX

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Project Overview

Objective

Analyze HR employee data to identify key drivers of employee attrition and provide actionable retention recommendations.

Tools Used

- Python – Pandas, Matplotlib, Seaborn, Scipy.
- Power BI – DAX, Tooltips, Smart Narrative.
- Statistics – Skewness, Correlation, Distribution

Dataset

- 1, 470 employees
- 35 columns
- IBM HR Analytics Dataset
- Source: Kaggle
- Data quality checks completed

Key Questions

- What is the attrition rate?
- Which department loses most?
- Is overtime associated with higher attrition?
- Does income affect attrition?
- Who is most likely to leave?

Goal: Identify WHY employees leave and HOW to retain them!

Methodology & Analytical Approach



- IBM HR Dataset
- Source: Kaggle
- 1,470 employees
- 35 features

- Checked missing values
- Fixed data types
- Removed duplicates
- Standardized formats

- Descriptive statistics
- Distribution analysis
- Correlation analysis
- Attrition analysis
- Outlier deter

- Python – Matplotlib
- Power BI – DAX Interactive dashboard
- Tooltip pages
- Smart Narrative

- Key findings
- Business impact
- Root cause analysis
- Action recommendation
- Retention strategies

Tools & Libraries used:

Python

Pandas

Matplotlib

Seaborn

Scipy

Power BI

DAX

Jupyter

Approach: Data-driven analysis combining statistical methods with visual storytelling

About the Dataset

1,470

Total Employees

35

Feature Columns

16.1%

Attrition Rate

Zero

Missing Values

Key Columns Analyzed

Age

Attrition

Department

Gender

JobRole

MonthlyIncome

OverTime

BusinessTravel

WorkLifeBalance

JobSatisfaction

YearsAtCompany

PerformanceRating

JobLevel

Data Source:
IBM HR Analytics Employee
Attrition Dataset.
Source: Kaggle.com

Descriptive Statistics

Statistical summary of key numeric columns

Column	Mean	Median	Min	Max	Distribution
Age	36.9	36.0	18	60	Normal
MonthlyIncome	6,503	4,919	1,009	19,999	Right Skewed
YearsAtCompany	7.0	5.0	0	40	Right Skewed
DailyRate	802.5	802.0	102	1,499	Normal
JobSatisfaction	2.7	3.0	1	4	Approximately symmetric
WorkLifeBalance	2.8	3.0	1	4	Left Skewed

Key Insight: MonthlyIncome is heavily right skewed (1.37) meaning most employees earn low salaries while a few top earners pull the average up significantly.

2

Normal
Distribution

2

Right Skewed
Distribution

1

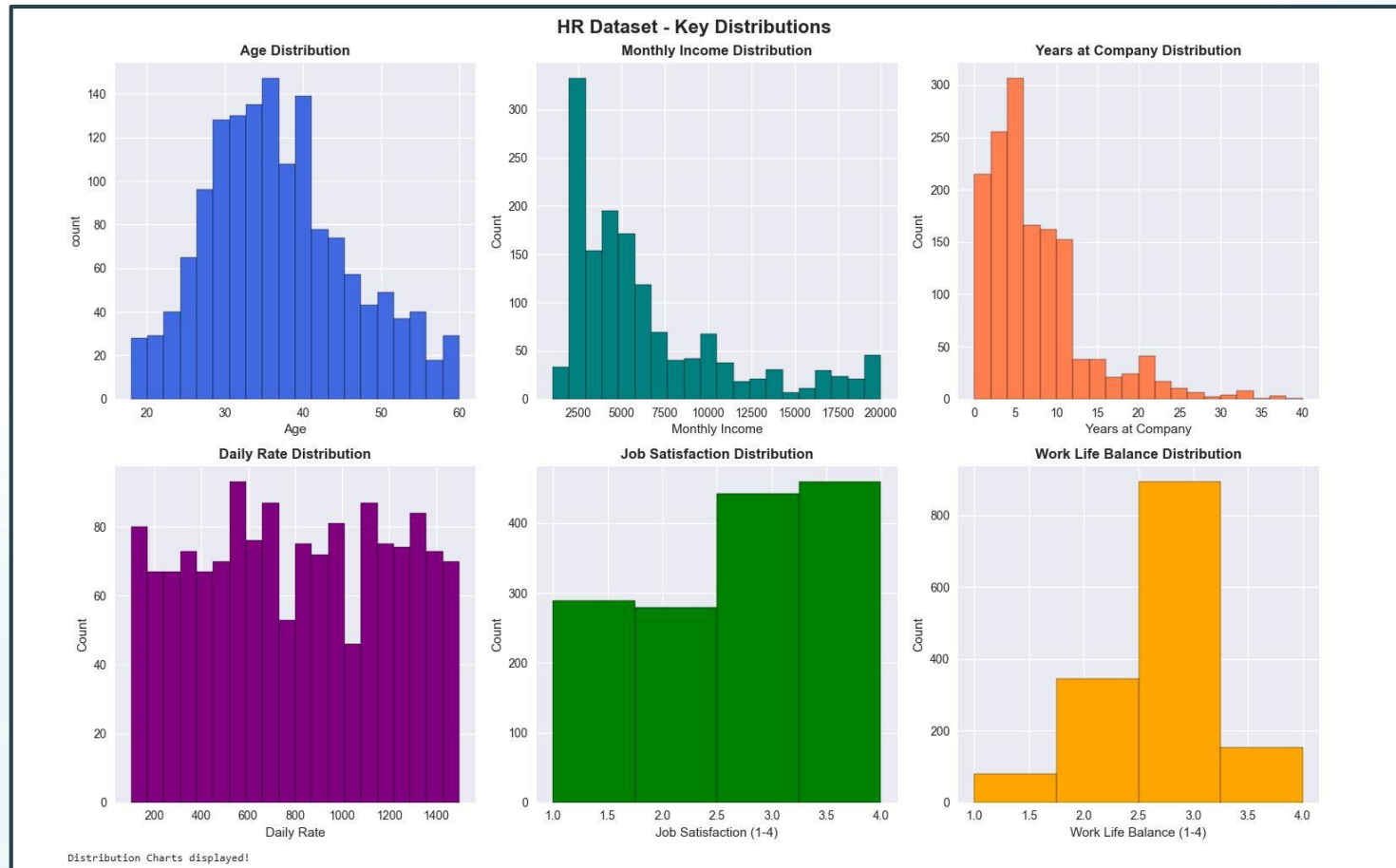
Left Skewed
Distribution

1

Approximately
symmetric

MonthlyIncome & YearsAtCompany are heavily right skewed – most employees fall within lower income bands!

Distribution Analysis



Age
Normal Distribution
Peak at 35-40 years

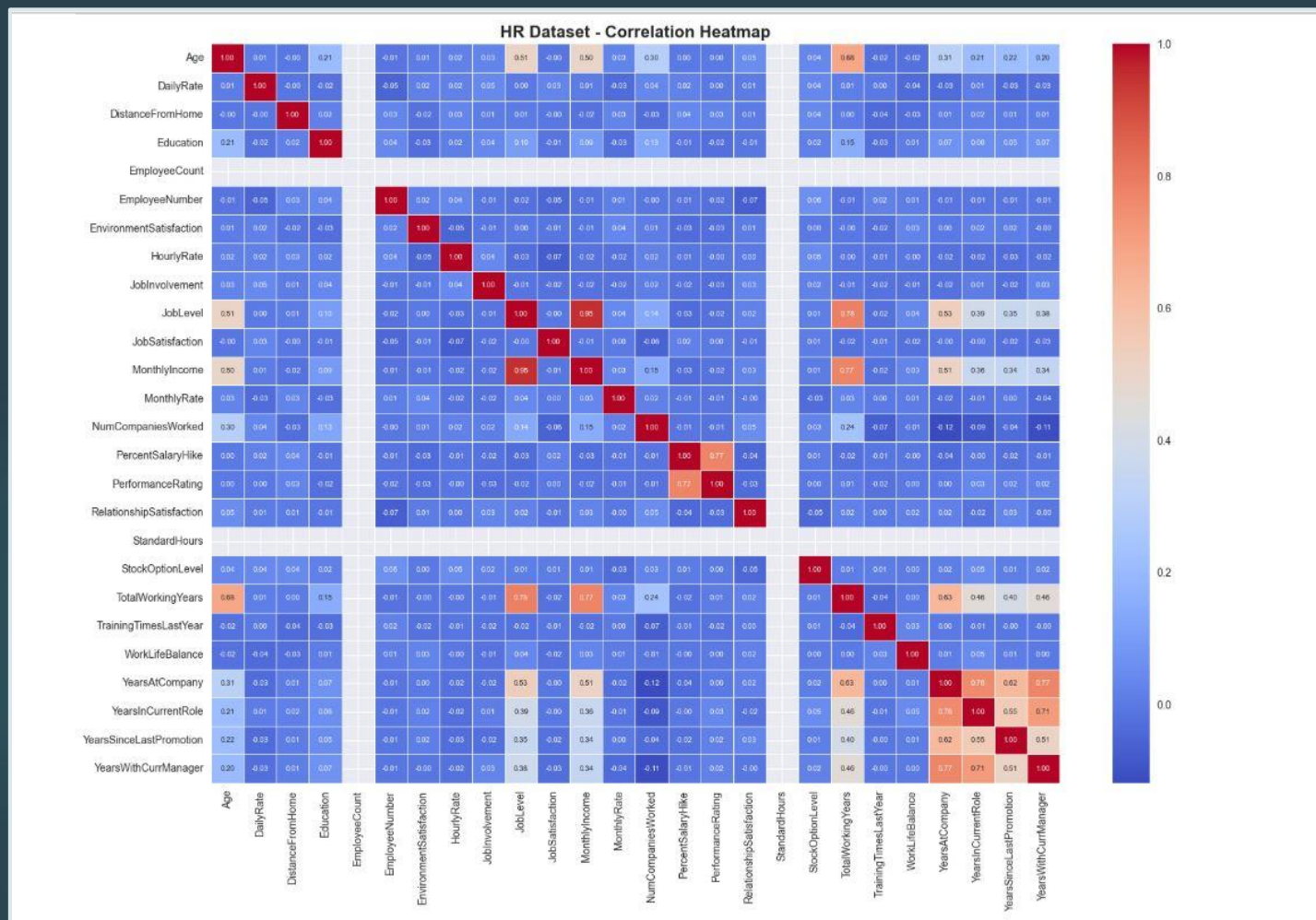
Monthly Income
Right Skewed
Most earn below
\$5k

Years at Company
Right Skewed
Most have under 5
years

Work Life Balance
Left Skewed
Most rate it highly

Key Finding: The workforce contains a high proportion of relatively new employees!

Correlation Analysis



Relationship	Value	Type
JobLevel & MonthlyIncome	0.95	● Strong+
JobLevel & TotalWorkingYears	0.78	● Strong+
PercentHike & Performance	0.77	● Strong+
MonthlyIncome & WorkingYears	0.77	● Strong+
YearsAtCompany & CurrentManager	0.77	● Strong+

Strongest Finding:
JobLevel & MonthlyIncome correlation of 0.95 means salary is strongly associated with job level!

Key Finding: Job Level shows the strongest relationship with salary among the variables analyzed.

Attrition Overview

1,470

Total Employees

1,233

Employees Stayed
83.9%

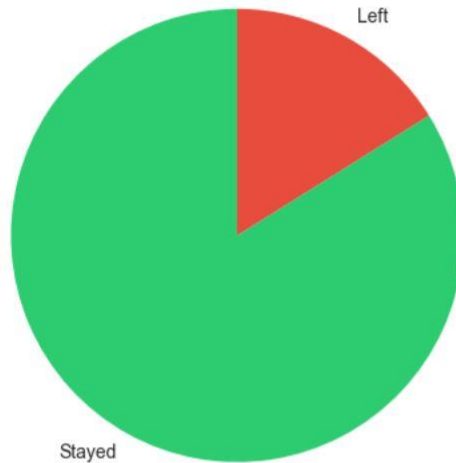
237

Employees Left
16.1%

10-12%

Industry Average
Attrition Rate

Employee Attrition Rate



<Figure size 800x550 with 0 Axes>
Attrition Overview displayed!

Attrition Count



WARNING

The attrition rate of 16.1% exceeds commonly reported industry benchmarks of 10-12% and this warrants further investigation.

Key Finding: The attrition rate is approximately 60% higher than the benchmark range used in this analysis.

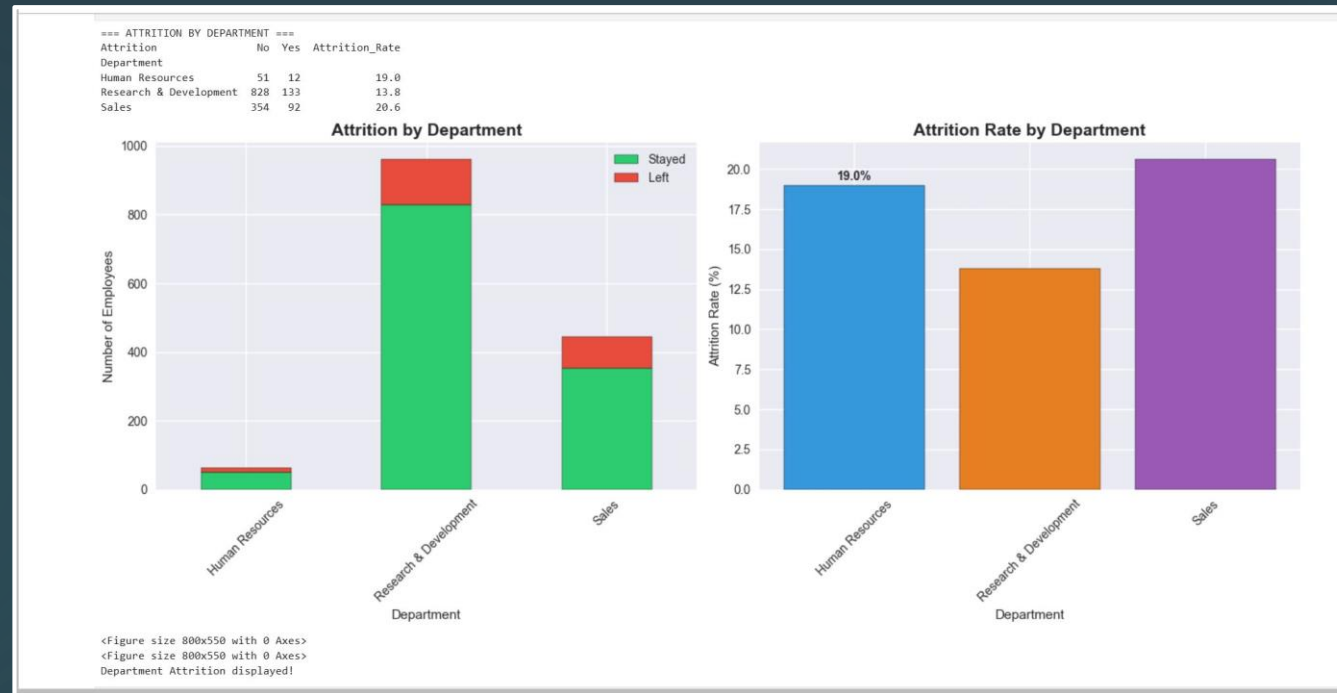
Department & Job Role Analysis

Sales
92 Left
20.6% Rate
Highest Rate

Human Resources
12 Left
19.0% Rate
High Risk

Research & Dev
133 Left
13.8% Rate
Lowest Rate

Remember: Count $\neq$ Rate.
R&D has highest COUNT
(133) but Sales has
highest RATE (20.6%)
because Sales is a smaller
department.



Top Job Roles by Attrition:

1. Sales Executive – 22.18% of all leavers.
2. Research Scientist – High count
3. Laboratory Tech – Significant numbers
4. Sales roles need urgent attention.

Key Finding: Sales has highest attrition RATE (20.6%) – 1 in 5 sales employees is leaving

Age & Income Analysis

37.6

Avg Age Stayed

33.6

Avg Age Left

\$6,833

Avg Income Stayed

\$4,787

Avg Income Left



Age Gap:
 $37.6 - 33.6 = 4$ years
Younger employees leaving faster.

Income Gap:
 $\$6,833 - \$4,787 = \$2,046$ per month, \$24,552 per year.

Employees who left tended to be younger and earned lower incomes than employees who stayed.

Key Finding: Employees who left earned \$24,552 LESS per year – salary is a critical retention factor.

Overtime & Travel Analysis

30.5%

Attrition WITH Overtime
Critical!

10.4%

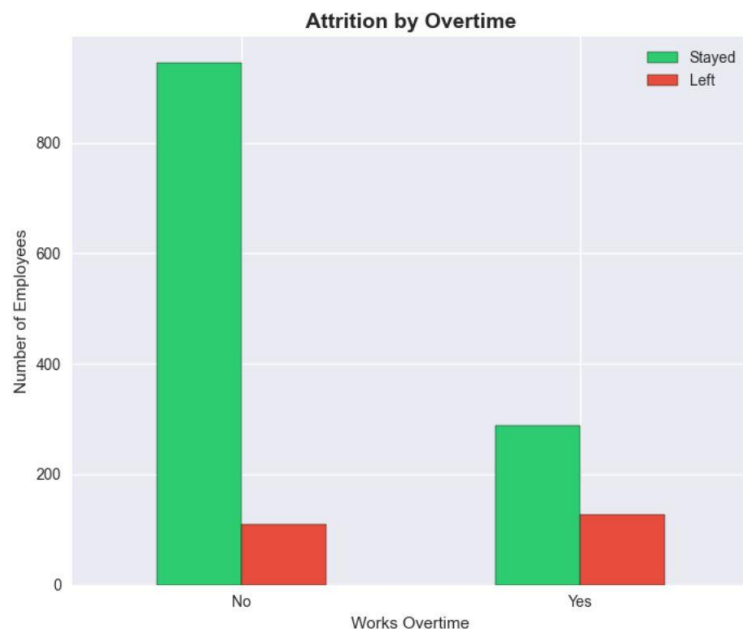
Attrition WITHOUT
Overtime
Manageable

24.9%

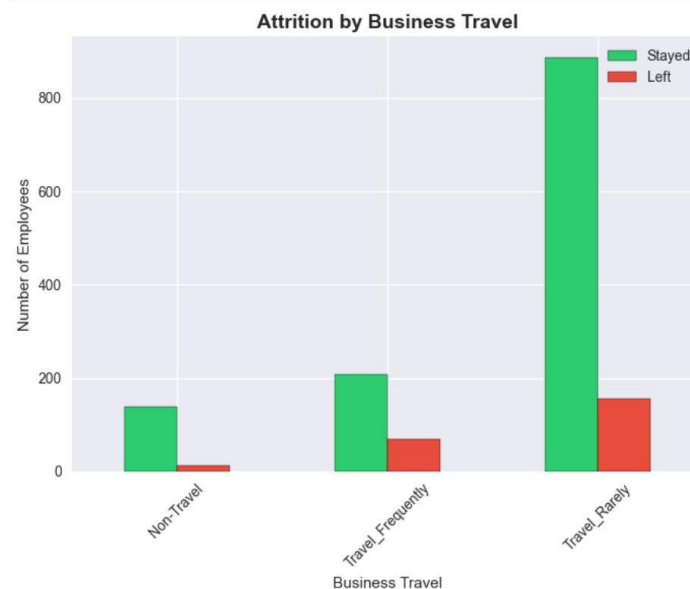
Travel Frequently
Attrition Rate
Highest

8.0%

Non Travel
Attrition Rate
Lowest



=== OVERTIME ATTRITION ANALYSIS ===
Attrition Rate WITH Overtime: 30.5%
Attrition Rate WITHOUT Overtime: 10.4%



=== BUSINESS TRAVEL ATTRITION ===
Non-Travel Attrition Rate: 8.0%
Travel_Frequently Attrition Rate: 24.9%
Travel_Rarely Attrition Rate: 15.0%

Risk Multiplier:
 $30.5\% / 10.4\% \approx 3X$ higher
attrition risk.

Business Travel:
 $24.9\% / 8.0\% = 3X$ higher attrition
risk.

Both factors
TRIPLE the risk of losing
employees!

Key Finding: Overtime & Frequent Travel TRIPLE attrition risk – should be prioritized for future investigation!

Work Life Balance

31.2%

Level 1
Low WLB
Critical

16.9%

Level 2
Medium WLB
High

14.2%

Level 3
Good WLB
Lowest

17.6%

Level 4
Excellent WLB
Interesting



=== WORK LIFE BALANCE ATTRITION ===

Level 1 (Low): 31.2%
Level 2 (Medium): 16.9%
Level 3 (Good): 14.2%
Level 4 (Excellent): 17.6%

Interesting Finding

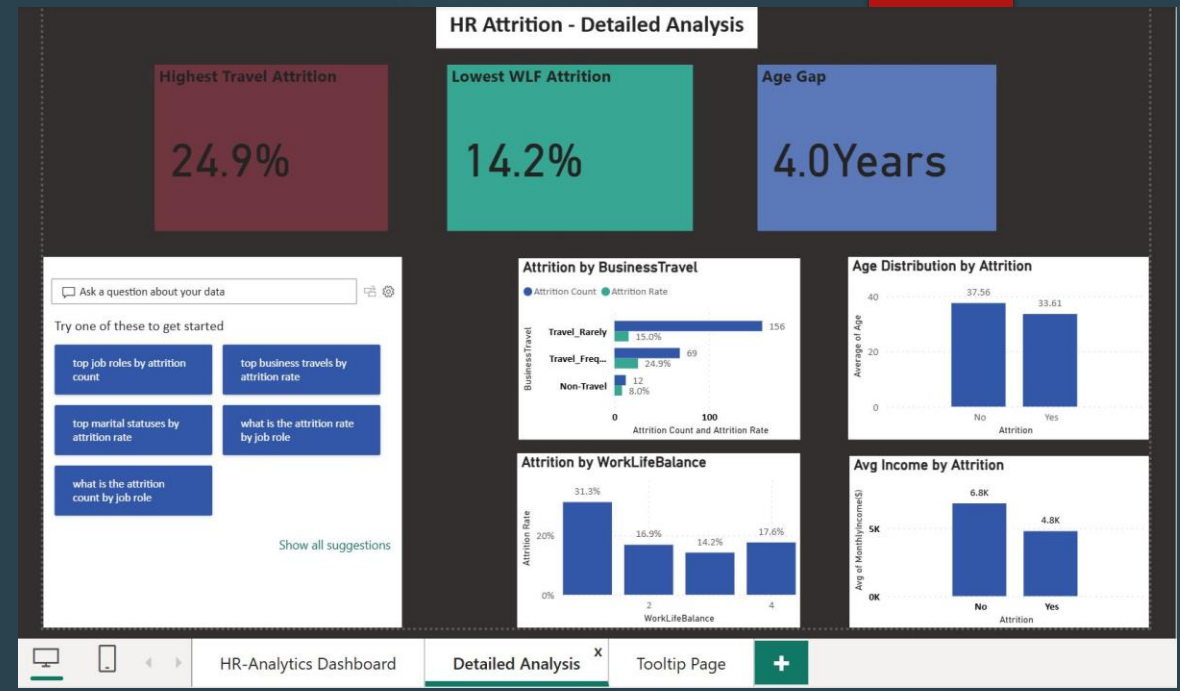
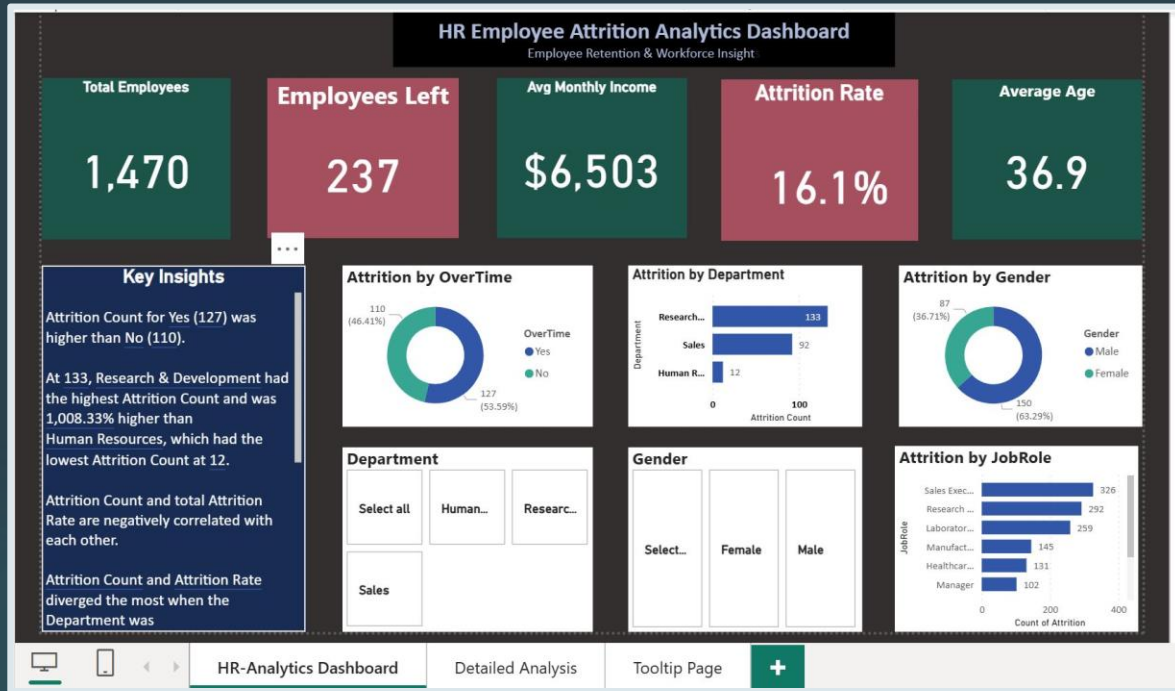
Level 4 (Excellent WLB) has HIGHER attrition than level 3 (Good WLB)

17.6% VS 14.2%

Why? Employees with excellent WLB have more energy & confidence to job hunt elsewhere.

Key Finding: Poor WLB (Level 1) has 31.2% attrition – more than DOUBLE the best level!

Power BI Dashboard Overview



Smart Narrative
DAX Measures
Dynamic Insights
Gender Filter
Tooltip Page
Interactive Slicers
Department Filter
Q&A Visual

Business Impact of Dashboard

1. Workforce visibility

Enables HR to track attrition trends in real time across departments and demographics

2. Risk identification

Highlights high-risk groups based on overtime, travel, income, and job role

3. Decision support

Supports targeted retention strategies for Sales, low-income employees, and high-overtime staff

4. Efficiency improvement

Reduces manual HR reporting through automated Power BI insights

Built with Power BI Desktop – Featuring Advanced DAX, Tooltips & Smart Narrative!

Key Findings & Recommendations

Key Findings

16.1% attrition rate – above industry average of 10-12%

Overtime triples attrition risk
30.5% vs 10.4%

Leavers earn \$2,040 less per month
(\$24,552 per year)

Younger employees leaving faster
avg age 33.6 vs 37.6

Sales highest rate
20.6% attrition

Poor WLB Level 1
31.2% attrition rate

Recommendations

Reduce overtime immediately
Assess workload distribution in departments with elevated attrition

Review salary structure
Conduct compensation benchmarking for roles with elevated attrition risk

Limit frequent travel
Use virtual meetings where possible

Focus on Sales department
Investigate workload & management quality

Create youth retention program with mentorship & clear career paths

Improve WLB policies.
Flexible hours & remote work options